Signal-to-Noise Ratio Detection in Educational Video Content: A Binary Classification Framework Using Transcript Embeddings

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Abstract

Educational content platforms face a structural misalignment: engagement metrics optimize for emotional resonance while learners seek informational value. We present a Signal-to-Noise Ratio (SNR) framework for automated educational content quality assessment applied to YouTube video transcripts. Our taxonomy defines HIGH signal content as material containing concrete actionable steps or methodologically sound explanations, and LOW signal content as material dominated by generic advice, fear-based framing, repetition, or promotional material.

We construct a dataset of 150 real YouTube transcripts (149 received usable ensemble labels; 1 excluded due to simultaneous API failures) and 300 synthetically generated transcripts spanning three domains: career and self-improvement, technology and artificial intelligence, and general education. Real transcripts are labeled via a three-model LLM ensemble (GPT-4o mini, Gemini 2.5 Flash, Llama 3.3 70B) using calibrated domain-specific rubrics, achieving 74% full three-model consensus and 82% average pairwise agreement—a significant improvement over the 53% pairwise agreement obtained in the initial domain-agnostic setup (which differed in both rubric and judge ensemble). Labels are treated as silver quality throughout.

A lightweight binary classifier is trained on 89 real transcripts mixed with 300 synthetic transcripts and evaluated on a held-out set of 60 real transcripts (89 + 60 = 149 usable real transcripts total). The best-performing SVM classifier achieves F1 = 0.871 with recall = 0.964 on the HIGH signal class (measured against silver labels; this reflects internal consistency with the labeling pipeline rather than human-validated accuracy), compared to a majority-class baseline of F1 = 0.000. We demonstrate that domain-specific rubric calibration and real-data augmentation improve both annotation reliability and classification performance relative to synthetic-only training baselines (synthetic-only SVM achieves F1 = 0.747 on the same test set).

All code, datasets, and generation prompts are publicly available at https://github.com/biditdas18/snr-detector.

1 Introduction

The proliferation of educational video content on platforms such as YouTube has created an information quality problem at scale. While over 500 hours of video are uploaded every minute [11], platform recommendation algorithms optimize primarily for watch time, click-through rate, and engagement—metrics that correlate with emotional stimulation rather than informational value. A fear-mongering video about job loss, a motivational montage of generic advice, or a heavily promoted course-selling video may outperform a methodologically rigorous tutorial on the same topic.

This misalignment imposes real costs on learners. Users investing time in low-signal content receive anxiety without actionability, motivation without method, and promotion without substance. The problem is particularly acute in high-stakes domains such as career development, technical skill-building, and immigration decisions, where poor information quality can lead to consequential decisions based on noise.

We propose a Signal-to-Noise Ratio (SNR) framework that formalizes content quality as a binary classification task: does this video transcript contain high-quality educational signal? The binary formulation is intentional. It maps directly to practical deployment scenarios—platform pre-recommendation filters, user-facing quality badges, and personalized content curation systems—where a yes/no quality signal is more actionable than a multi-tier rating.

The core challenge is constructing a quality classifier that is (1) grounded in a principled taxonomy, (2) trained without exhaustive manual annotation, (3) deployable at inference speeds impractical for frontier LLMs, and (4) honest about its limitations at this stage of development.

This paper makes the following contributions:

1. A formal SNR taxonomy for educational video content with explicit criteria for HIGH and LOW signal classification.
2. A scalable dataset construction methodology combining LLM ensemble annotation for real transcripts and structured prompt-based generation for synthetic training data.
3. A lightweight binary classifier using transcript embeddings that approximates LLM judgment at a fraction of the inference cost.
4. A public dataset and codebase enabling reproducibility and community extension.

2 Related Work

Content Quality Assessment. Prior work on automated content quality has focused primarily on news credibility [1], fact verification [2], and readability scoring [3]. Educational content quality specifically remains underexplored, with most work addressing either MOOCs with structured curricula or child-directed content with platform-specific metadata. YouTube educational content presents a distinct challenge: creator-defined structure, unverified claims, and engagement-optimized framing exist alongside genuinely high-value instructional material.

LLM-as-Judge. Recent work has established the viability of using large language models as automated annotators. Zheng et al. [4] demonstrate that GPT-4 judgments on open-ended question answering correlate strongly with human preference, introducing the MT-Bench benchmark. Fu et al. [5] show that LLM scoring can be steered toward arbitrary evaluation dimensions through structured prompting. We build on this paradigm, using a three-model ensemble to reduce single-model bias.

Knowledge Distillation. Our approach is conceptually related to knowledge distillation [6], where a lightweight student model is trained to approximate the behavior of a more capable teacher. Here, the LLM ensemble serves as the teacher, generating silver labels on real transcripts, while the embedding classifier serves as a deployable student.

Sentence Embeddings. We use sentence-transformer models for transcript representation [7]. These models have demonstrated strong performance on semantic similarity and classification tasks, and their compact representations enable efficient inference in resource-constrained deployment settings.

3 SNR Taxonomy

We define a binary taxonomy for educational video content quality based on the information value delivered to a learner per unit of content consumed.

3.1 HIGH Signal Content

A transcript is classified as HIGH signal if it satisfies at least two of the following criteria:

- **Methodological coherence:** Ideas build logically on each other. The transcript is not a random assembly of disconnected tips.
- **Specificity:** Content includes concrete actionable steps, named frameworks, specific statistics, or examples grounded in verifiable reality.
- **Informational novelty:** The transcript provides perspective or knowledge the viewer likely did not already possess.
- **Solution orientation:** When addressing problems or risks, the transcript provides analysis of causes and suggested next steps, not merely alarming framing.

3.2 LOW Signal Content

A transcript is classified as LOW signal if it exhibits one or more of the following patterns at a dominant level (>30% of content):

- **Generic advice:** Vague recommendations without specificity (e.g., "work hard," "believe in yourself," "be consistent") unsupported by method or example.
- **Repetition without progression:** The same point restated in different words with no new information added.
- **Fear-based framing:** Emotional urgency or outrage used as the primary hook, with no actionable solution offered.
- **Promotional content:** Course selling, channel promotion, or product placement dominating the substance.
- **Logical fallacies:** Appeal to authority without evidence, manufactured urgency, social pressure, or burden-of-proof shifting.

3.3 Conflict Resolution

When a transcript contains HIGH-signal content alongside dominant LOW-signal patterns, it is classified as LOW. This conservative policy reflects deployment intent: a platform filter should not surface content where significant portions undermine the signal value.

4 Dataset Construction

Our dataset comprises three components: a real-transcript evaluation set, a real-transcript training supplement, and a synthetic training set.

4.1 Real Transcript Dataset (150 Transcripts)

We collected 150 YouTube video transcripts across three domains: career and self-improvement, technology and artificial intelligence, and general education. Transcripts were obtained programmatically using the `youtube-transcript-api` library, requiring no manual video viewing.

Of the 150 collected transcripts, 149 received usable ensemble labels (1 excluded due to simultaneous API failures from two of three models). These are split into two subsets: **Evaluation set** (60 transcripts): Twenty per domain, used exclusively as the held-out test set. Their labels were committed to GitHub before any model training, providing a timestamped integrity record. **Training supplement** (90 collected; 89 retained after the one exclusion): Thirty per domain collected from established educational channels including IBM Technology, ByteByteGo, TED-Ed, and CrashCourse. These are mixed with synthetic data during training to address synthetic-to-real domain shift. The arithmetic 89 + 60 = 149 usable transcripts is consistent throughout.

Rubric Calibration. Initial annotation using a single domain-agnostic rubric with a three-model ensemble (GPT-4, Gemini 1.5 Pro, Meta AI) yielded pairwise inter-model agreement of 53%, approaching the 50% random baseline for binary classification. We identified two root causes: (1) content quality criteria are domain-dependent—actionability in career advice requires procedural steps and timelines, while actionability in general education requires conceptual clarity rather than action items; and (2) individual models exhibit systematic labeling biases that a shared rubric cannot resolve.

To address these issues, we developed calibrated domain-specific rubrics through an iterative multi-agent process. Three judge models (Llama 3.1 8B, Mistral 7B, Qwen 2.5 7B) independently evaluated each transcript; a coordinator model (Claude Sonnet) diagnosed disagreement patterns from per-model criterion quotes and proposed targeted rubric revisions. This loop ran until weighted inter-judge agreement reached convergence or stagnation. The process converged with final rubrics achieving strong calibration validated against the production labeling ensemble (see Appendix A).

LLM Ensemble Annotation. All 150 transcripts were labeled using a consistent three-model ensemble: GPT-4o mini (OpenAI), Gemini 2.5 Flash (Google), and Llama 3.3 70B (Groq), each receiving identical calibrated domain-specific rubrics. This approach follows the LLM-as-judge paradigm [4], reducing single-model bias through ensemble aggregation. Final labels were determined by majority vote; one transcript (0.7%) was excluded due to simultaneous API failures from two of three models.

The calibrated domain-specific rubrics improved average pairwise inter-model agreement from 53% (domain-agnostic) to 82% across domains, with 74% of transcripts achieving full three-model consensus. Weighted inter-judge agreement—assigning 1.0 for unanimous consensus and 0.667 for majority consensus—reached 91% overall. All labels are treated as silver quality; human validation is planned as future work.

Why LLM Annotation Over Human Annotation. Human annotation of educational content quality is subject to two systematic biases that motivated our LLM ensemble design:

Creator bias occurs when annotator familiarity with a channel contaminates content-level judgment. A viewer who associates a creator with promotional or fear-based content will apply that reputation to every transcript regardless of what the content actually contains. Conversely, a viewer who trusts a creator unconditionally may rate low-signal content as HIGH. In both cases, the label reflects creator identity rather than transcript quality.

Domain bias occurs when annotator interest in a subject inflates or deflates perceived signal. A software engineer may rate lifestyle advice as inherently noisy regardless of content structure. A viewer deeply invested in immigration content may rate general geopolitics discussions as LOW signal by personal preference, not by rubric. Domain bias means that annotation quality is bounded by the annotator's subject matter coverage—an unacceptable constraint for a framework intended to generalize across domains.

LLM ensemble annotation mitigates both failure modes. Language model annotation can reduce creator- and domain-driven human annotator bias, particularly when creator identity is withheld, producing labels that more closely reflect content structure—though it does not eliminate model-level bias inherited from pretraining, which is why we treat these labels as silver quality. This property is particularly valuable for platform-scale deployment, where quality labels must generalize across creator reputations and viewer demographics.

The evaluation set contains 28 HIGH (47%) and 32 LOW (53%) transcripts. Table 1 shows the evaluation set distribution by domain.

Table 1: Real Transcript Evaluation Set Distribution (n=60)

Domain	HIGH	LOW	Total
Career & Self-Improvement	4	16	20
Technology & AI	7	13	20
General Education	17	3	20
Total	28	32	60

4.2 Synthetic Training Set

To enable balanced classifier training without manual annotation, we generated 300 synthetic transcripts using the Anthropic Claude API. Each transcript was generated using a structured domain-specific prompt explicitly designed to elicit either HIGH or LOW signal content (see Appendix B for full prompt templates).

Labels for synthetic transcripts are assigned by construction: a transcript generated by a HIGH-signal prompt is labeled HIGH by definition. This removes manual annotation ambiguity for the synthetic subset, though it does not guarantee that every generated transcript perfectly satisfies its intended class definition—prompt-adherence errors remain possible. We therefore treat synthetic labels as high-confidence but not error-free.

The synthetic training set comprises 150 HIGH and 150 LOW transcripts distributed equally across the three domains (50 HIGH + 50 LOW per domain), yielding a balanced training set that mitigates class imbalance during learning. Synthetic transcripts target 400–500 words to approximate the length distribution of real transcripts in our evaluation set.

Table 2: Synthetic Training Set Distribution

Domain	HIGH	LOW	Total
Career & Self-Improvement	50	50	100
Technology & AI	50	50	100
General Education	50	50	100
Total	150	150	300

4.3 Dataset Integrity Protocol

All label files (`test_set_final.csv`, `train_set_final.csv`, `all_150_silver_labels.csv`) and result artifacts are version-controlled in a public GitHub repository, providing an immutable, publicly auditable record. Labels were not modified after publication. Synthetic transcript labels are deterministic by construction and fully reproducible from the published generation prompts.

5 Methodology

Figure 1 illustrates the full SNR-Detect system. The annotation pipeline (top) runs offline to produce silver labels; the classification pipeline (bottom) runs at inference on any YouTube transcript.

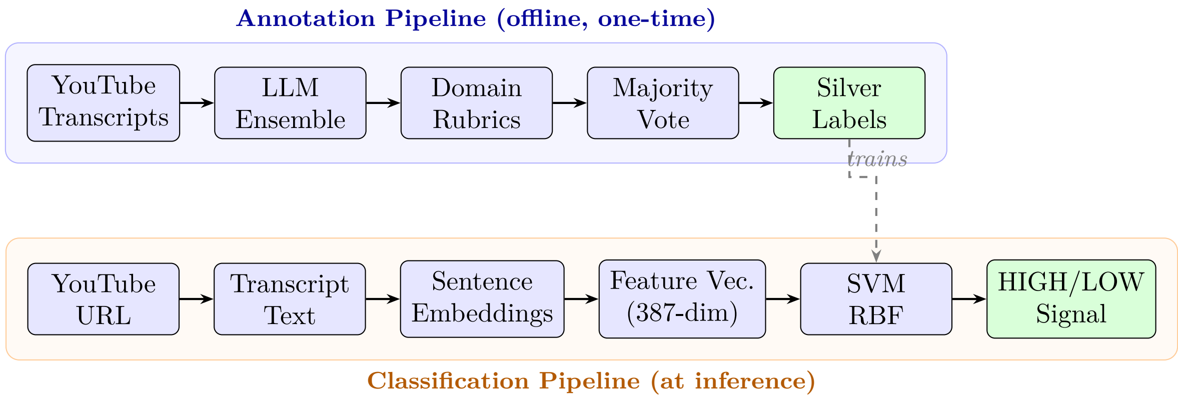

Figure 1: SNR-Detect system architecture. *Top:* The annotation pipeline labels real transcripts offline using a three-model LLM ensemble with calibrated domain-specific rubrics. *Bottom:* The classification pipeline embeds any YouTube transcript into a 387-dimensional feature vector (384 sentence embedding + 3 domain one-hot) and classifies it as HIGH or LOW signal using a trained SVM classifier. The dashed arrow indicates that silver labels from the annotation pipeline are used to train the classifier.

5.1 Task Formulation

We formulate SNR classification as binary: given a YouTube video transcript t, predict $y \in \{\text{HIGH}, \text{LOW}\}$. HIGH indicates content worth watching for its educational value; LOW indicates content that fails to deliver meaningful signal relative to time invested.

5.2 Transcript Representation

We represent transcripts as dense semantic embeddings using the `all-MiniLM-L6-v2` sentence-transformer model [7], mapping variable-length transcript text to fixed 384-dimensional vectors. To provide domain context without splitting the classifier, we concatenate a three-dimensional one-hot domain encoding, yielding a 387-dimensional feature vector per transcript. This allows the classifier to learn domain-specific decision boundaries within a single unified model while preserving the full training dataset size.

Embeddings are computed independently for training and test transcripts, with no information leakage between splits.

5.3 Classifiers

We train two binary classifiers on the mixed training set:

- **Logistic Regression (LR):** L2 regularization, C=1.0, `class_weight='balanced'`, trained to convergence (`max_iter=1000`).
- **Support Vector Machine (SVM):** RBF kernel, C=1.0, `class_weight='balanced'`, selected for effectiveness in high-dimensional embedding spaces [8].

Both classifiers are trained on a mixed set of 389 transcripts (89 real silver-labeled + 300 balanced synthetic) and evaluated on the 60 held-out real transcripts. The mixed training set directly exposes the classifier to real YouTube content distributions during training, addressing the synthetic-to-real domain shift observed in synthetic-only baselines.

5.4 Baseline

We compare against a majority-class baseline that predicts LOW for every transcript. This baseline represents the trivially achievable performance ceiling for a zero-information classifier and demonstrates that accuracy alone is insufficient for evaluating imbalanced classification tasks [9].

5.5 Evaluation Protocol

We evaluate on the 60-transcript real silver-labeled test set using:

- **Accuracy:** overall classification rate
- **F1 (HIGH class):** primary metric given class imbalance; harmonic mean of precision and recall for the minority HIGH class
- **Precision and Recall:** breakdown of HIGH-class performance
- **Confusion matrix:** full outcome breakdown

F1 on the HIGH class is the primary metric, as it captures the classifier's ability to identify genuine educational signal. The calibrated domain-specific rubric produced a more balanced test set (47% HIGH, 53% LOW) than initial annotation (20% HIGH), reflecting the rubric's improved ability to distinguish genuinely high-signal content from borderline cases.

6 Results

6.1 Classification Performance

Table 3 presents classification results on the 60-transcript held-out evaluation set.

Table 3: Classification Results on Real Transcript Test Set (n=60)

Model	Acc.	F1	Prec.	Rec.
Baseline (majority LOW)	0.533	0.000	0.000	0.000
Synthetic-only training (300 synthetic transcripts; matched test set)				
SVM RBF (synth-only, thresh=0.50)	0.683	0.747	0.596	1.000
Log. Regression (synth-only, thresh=0.50)	0.683	0.732	0.605	0.929
Mixed training (89 real + 300 synthetic; 389 total)				
Log. Regression (thresh=0.55)	0.850	0.857	0.771	0.964
SVM RBF (thresh=0.50)	0.867	0.871	0.794	0.964

The SVM classifier achieves F1 = 0.871 on the HIGH signal class, identifying 96.4% of genuine HIGH signal transcripts (recall = 0.964). The majority-class baseline achieves only 0.533 accuracy—confirming that the calibrated rubric produced a near-balanced test set and that accuracy alone is an insufficient metric for this task. The trained classifier provides strong discriminative signal where the baseline provides none.

The high recall (0.964) reflects a deliberate threshold choice appropriate for content filtering: surfacing a false positive (mislabeled LOW as HIGH) is preferable to missing genuine educational signal. Table 3 shows directly comparable synthetic-only baselines trained and evaluated on the same splits. The mixed-training SVM improves F1 from 0.747 (synthetic-only) to 0.871 and precision from 0.596 to 0.794, demonstrating that mixing 89 real transcripts into training substantially reduces the false positive rate by exposing the classifier to authentic YouTube content distributions during learning.

6.2 Domain Feature Ablation

To assess whether the classifier exploits domain priors encoded in the 3-dimensional one-hot domain feature, we trained an SVM-RBF on the full mixed training set using 384-dimensional sentence embeddings only (no domain one-hot). The no-domain SVM achieves F1 = 0.844, accuracy = 0.833, precision = 0.750, and recall = 0.964 on the held-out test set (results saved to `reports/ablation_no_domain.json`). The modest F1 reduction of 0.027 relative to the full 387-dimensional model (F1 = 0.871) suggests the domain feature provides marginal discriminative value beyond what the semantic embedding already captures, and that the classifier is not primarily exploiting domain-level class imbalance.

6.3 Annotation Reliability

The calibrated domain-specific rubrics improved average pairwise inter-model agreement from 53% (initial domain-agnostic rubric) to 82% across career and general education domains, with the technology/AI domain achieving 95% pairwise agreement between GPT-4o mini and Gemini 2.5 Flash. Weighted inter-judge agreement reached 91% overall (74% full three-model consensus). Because the initial (53%) and production (82%) settings differ in both the rubric and the judge ensemble (GPT-4 / Gemini 1.5 Pro / Meta AI vs. GPT-4o mini / Gemini 2.5 Flash / Llama 3.3 70B), this improvement reflects their combined effect; isolating the causal contribution of rubric calibration alone would require holding the judge ensemble fixed across both rubrics. The 82% figure refers to the production frontier ensemble; the local calibration panel reported in the companion paper converged on two of three domains, with the Career domain remaining below target.

The test set labels were assigned by the same class of frontier LLM models used to label the training supplement, which may introduce correlated label bias. Human validation remains planned future work.

6.4 Confusion Matrix

Figure 2 presents the confusion matrix for the best-performing classifier (SVM RBF, threshold = 0.50) on the held-out test set.

6.5 Error Analysis

We inspected misclassified transcripts to characterize systematic failure modes. Two residual patterns emerged:

- **Motivational framing with embedded specifics:** Transcripts containing genuine actionable content but opening with emotional hooks that activate LOW-signal features in embedding space. Career self-improvement content is particularly susceptible, as motivational language and procedural steps co-occur more frequently in this domain.
- **Boundary ambiguity:** Transcripts with roughly equal proportions of HIGH and LOW signal content present irreducible classification difficulty. The 82% average pairwise

		Predicted Label	
		Predicted LOW	**Predicted HIGH**
True Label	**True LOW**	25	7
	True HIGH	1	27

Figure 2: Confusion matrix on the 60-transcript held-out test set (SVM RBF, threshold = 0.50). Rows indicate true labels; columns indicate predicted labels. Train set: 89 real + 300 synthetic transcripts. The classifier achieves recall = 0.964 on the HIGH signal class.

inter-model agreement confirms that human-like boundary cases are not unique to the classifier—even frontier LLMs disagree on these examples.

7 Limitations

Silver label quality. All labels in this work are silver quality, assigned by LLM ensemble majority vote following calibrated domain-specific rubrics. While inter-model agreement improved substantially from 53% (domain-agnostic) to 82% average pairwise (domain-specific), LLM judges may reflect systematic biases inherited from pretraining. The test set labels were produced by the same class of frontier LLM models used to label the training supplement, which may introduce correlated label bias. Human validation via crowdsourced annotation with Cohen's κ computation is planned as immediate future work.

Real training data concentration. The 89 real training transcripts were collected predominantly from established educational channels (IBM Technology, ByteByteGo, TED-Ed, CrashCourse), which skew toward HIGH signal content. LOW signal representation in training is primarily provided by synthetic transcripts, which may not fully capture naturalistic LOW signal patterns.

YouTube scope. This study is scoped exclusively to YouTube transcripts. The SNR framework may not transfer without adaptation to shorter-form content (e.g., Instagram Reels, TikTok) where transcript length and narrative structure differ substantially.

Narrow domain coverage. Three domains were selected for this pilot study. Generalization to medical, legal, and financial educational content requires additional validation given the higher stakes of quality errors in those domains.

Small real test set. The evaluation set contains 60 transcripts (n=60), limiting the statistical reliability of reported metrics. Given the small test set, we report 95% Wilson score intervals for the rate-based metrics, computed from the confusion matrix: accuracy 0.867 [0.758, 0.931], recall 0.964 [0.823, 0.994], precision 0.794 [0.632, 0.897]. These wide intervals underscore that all reported metrics are feasibility-stage estimates rather than stable performance claims; a bootstrap interval for F1 and larger-scale validation are scoped to future work. This paper positions itself as a feasibility study and framework contribution.

8 Future Work

Human validation. We will recruit independent annotators via Prolific to label a transcript subset, enabling computation of human-LLM agreement (Cohen's κ) as a more reliable evaluation benchmark. Annotators will be selected across domains to measure whether domain bias affects human label quality relative to the LLM ensemble baseline.

Two-Mode Deployment Architecture. We envision two distinct deployment modes with fundamentally different bias profiles:

Personal deployment treats annotator bias as a feature rather than a flaw. A user's domain preferences and creator associations are valid personal signal. A personal SNR tool learns from individual feedback, surfacing content that matches the user's definition of educational value and warning against content that matches their historical noise patterns. Human bias is intentionally preserved and personalized.

Platform deployment requires the opposite approach. A recommendation filter must generalize across millions of viewers with heterogeneous domain preferences and creator associations. Here, LLM ensemble annotation—and eventually, majority consensus from large-scale user feedback—washes out individual bias. Platform-scale labels reflect population-level signal quality, not any single annotator's preferences. Existing "was this helpful" user signals from platforms such as YouTube represent an untapped training signal for this mode.

This two-mode architecture cleanly separates the bias question: personal tools embrace it, platform tools reduce it. Both are valid applications of the SNR framework with different annotation strategies.

Personalized quality learning. A user-facing deployment can collect binary helpfulness feedback and periodically fine-tune a personal SNR classifier. Population Stability Index monitoring [10] detects preference drift and triggers model refresh, enabling adaptation as user educational interests evolve.

Domain expansion. Future dataset collection will extend to medical, legal, and financial educational content, where signal-to-noise distinction carries higher consequential stakes.

9 Conclusion

We presented SNR-Detect, a binary classification framework for educational video content quality assessment. Our SNR taxonomy formalizes the distinction between content that delivers genuine educational value and content dominated by noise patterns—a distinction that existing platform metrics fail to capture. We showed that moving to domain-specific rubrics together with the production labeling ensemble raised pairwise inter-model agreement from 53% to 82%, and that mixing real transcripts into classifier training improves performance, achieving F1 = 0.871 (measured against silver labels; reflecting internal consistency with the labeling pipeline rather than human-validated accuracy) with recall = 0.964 on the HIGH signal class—compared to F1 = 0.747 for a synthetic-only baseline evaluated on the same test set.

The key contributions are methodological: a reproducible pipeline for dataset construction without exhaustive manual annotation, a domain-specific rubric calibration procedure that substantially improves LLM ensemble annotation reliability, and a deployable classifier that approximates LLM judgment at a fraction of the inference cost. The framework lays groundwork for platform-level content filtering and personalized quality learning systems where real-time, low-cost inference is a hard requirement.

We release all code, datasets, and generation prompts at `https://github.com/biditdas18/snr-detector`.

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A Calibrated Domain-Specific Rubrics

The following calibrated rubrics were used for all 150 real transcript labels. Each domain received a separate rubric following iterative multi-agent calibration. All three annotators (GPT-4o mini, Gemini 2.5 Flash, Llama 3.3 70B) received identical domain-specific rubrics.

Career & Self-Improvement — HIGH SIGNAL requires at least 2 of: (1) Specific procedural steps with named methods, tools, timelines, or defined techniques; (2) Concrete outcomes or metrics—numbers, percentages, named frameworks; (3) Specific action plan or named solution when addressing a problem; (4) Sequential ideas where each point adds new information.

Career & Self-Improvement — LOW SIGNAL (any one dominant): (1) Motivational language dominates with no actionable method—phrases like "believe in yourself" appear repeatedly and procedural steps are absent; (2) Same warning or point repeated more than twice; (3) Primary hook is fear or urgency with no concrete mitigation; (4) More than 30% promotes a course, service, or product; (5) Claims are authoritative but cite no evidence—named books, frameworks, or cited statistics do NOT trigger this criterion.

Technology & AI — HIGH SIGNAL requires at least 2 of: (1) Named technologies, algorithms, or architectural patterns with accurate technical detail; (2) Mechanism or tradeoff explained—not just what but why; (3) Concrete and implementable code concepts, benchmarks, or design decisions; (4) Specific solution path for a technical problem.

Technology & AI — LOW SIGNAL (any one dominant): (1) Hype language without technical substance; (2) Sweeping unsubstantiated claims about AI capabilities or job market; (3) Primary message is fear with no actionable technical skill; (4) Technology recommended without explaining what it does or why; (5) More than 30% promotes a paid resource—a brief sponsor mention alone does NOT qualify.

General Education — HIGH SIGNAL requires at least 2 of: (1) Mechanism or concept explained from first principles with logical steps; (2) Specific analogy, thought experiment, or concrete example that illuminates the concept accurately; (3) Non-obvious insight or counter-intuitive conclusion; (4) Accurate representation of complexity; (5) Historical context, causal analysis, or named expert perspectives when covering problems.

General Education — LOW SIGNAL (any one dominant): (1) Alarming or conspiratorial framing as primary hook without evidence; (2) Confident causal claims without evidence or logical reasoning; (3) Complex phenomena presented as simple good-vs-evil narratives; (4) Emotional manipulation dominates over explanation. Note: absence of actionable steps does NOT indicate LOW signal for general education.

Conflict rule (all domains): If HIGH-signal content coexists with dominant LOW-signal patterns (>30% of transcript), assign LOW.

B Synthetic Training Transcripts

The 300 synthetic transcripts used for classifier training are available in the project repository at: `https://github.com/biditdas18/snr-detector/blob/main/data/synthetic/synthetic_transcripts.csv`

Each transcript includes domain, signal level (HIGH/LOW), and generated text. Labels are assigned by construction from the generation prompt—HIGH prompts produce HIGH transcripts by definition.